

KOM4521 Multivariable Control Theory

2024-2025 Fall

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Grading Policy:

- **%30 Midterm Exam**
- **%30 Homework (2 x %15)**
- **%40 Final Exam**

Syllabus:

Week 1-2-3

- Feedback Control Systems
- Mathematical Preliminaries
(Norm, Matrix Theory, Singular Value Decomposition, Cayley-Hamilton Theorem)
- State-Space Representation and Canonical Forms

Syllabus:

Week 4-5-6

- Diagonalization, Solution of LTI Systems
- Stability of LTI Systems, Lyapunov Stability Theorem
- Controllability and Observability

Syllabus:

Week 7-9-10

- Calculation of Controllability and Observability Grammians
- PBH Test for Controllability and Observability

Syllabus:

Week 11-12-13

- State Feedback Controller Design
- Observer Design
- Applications of State Feedback Controllers and Observers

Syllabus:

Week 14-15

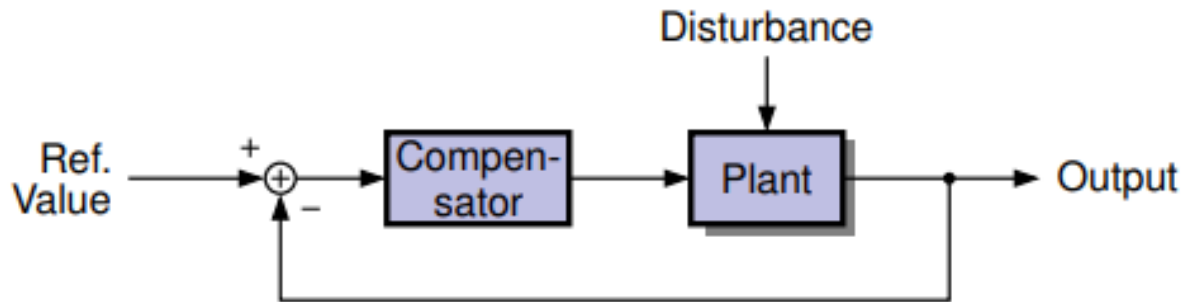
- Linear Quadratic Regulator
- Feedforward Controller Design

Feedback Control Systems:

Control is the process of causing a system variable to conform to some desired value, called a reference value (e.g., variable = temperature for a climate-control system).

Feedback is the process of measuring the controlled variable and using that information to influence its value.

Feedback Control Systems:



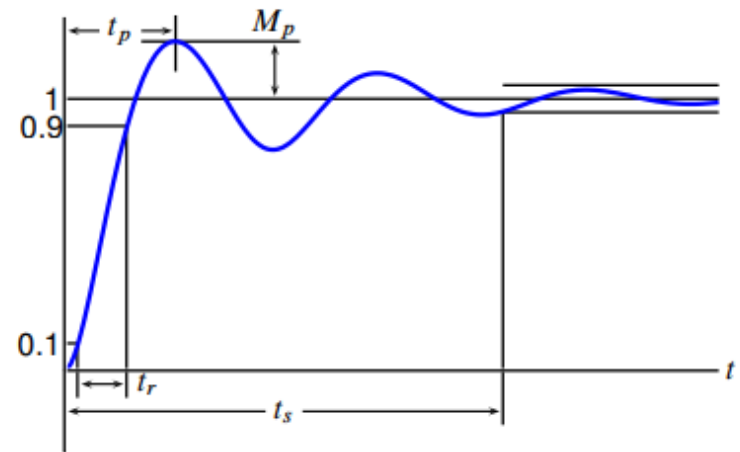
Feedback is not necessary for control. But, it is necessary to cater for system uncertainty, which is the principal role of feedback.

Open-loop control is also possible.

Feedback Control Systems:

Goals of feedback control:

- Change dynamic response of a system to have desired properties.
- Output of system tracks reference input.
- Reject disturbances.



Feedback Control Systems:

Multivariable, state-space control instead:

- Primarily uses time-domain matrix representations of systems.
- Very powerful. Can often place poles of closed-loop system anywhere we want!
Can make fast, smooth, etc.
- Same methods work for single-input, single-output (SISO) or multi-input, multi-output (MIMO) systems.

Mathematical Preliminaries:

NORM: A measure of length or gain.

- Properties of a vector norm, where x, y are vectors and α is scalar.

$$\|x\| \geq 0$$

$$\|x\| = 0 \quad \text{iff } x = 0$$

$$\|\alpha x\| = |\alpha| \|x\|$$

$$\|x + y\| \leq \|x\| + \|y\| .$$

Mathematical Preliminaries:

Vector p -norms: The p -norm of a vector is defined as

$$\|x\|_p = \lim_{\alpha \rightarrow p} \sqrt[\alpha]{\sum_{k=1}^{n_x} |x_k|^\alpha}.$$

The (Euclidean) 2-norm measures length of vector (from origin).

$$\|x\| = \|x\|_2 = \sqrt{x_1^2 + \cdots + x_n^2} = \sqrt{x^T x}.$$

Mathematical Preliminaries:

The ∞ -norm:

$$\|x\|_{\infty} = \lim_{\alpha \rightarrow \infty} \sqrt[\alpha]{\sum_{k=1}^{n_x} |x_k|^{\alpha}} = \max_k |x_k|.$$

The 1-norm:

$$\|x\|_1 = \lim_{\alpha \rightarrow 1} \sqrt[\alpha]{\sum_{k=1}^{n_x} |x_k|^{\alpha}} = \sum_{k=1}^{n_x} |x_k|.$$

Mathematical Preliminaries:

Quadratic forms: A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ of the form

$$f(x) = x^T A x = \sum_{i,j}^n A_{ij} x_i x_j$$

is called a quadratic form. In a quadratic form, we may assume A is symmetric since

$y = y^T$, so
$$y = x^T A x = x^T A^T x = x^T ((A + A^T)/2) x,$$

where $(A + A^T)/2$ is called the symmetric part of A .

Mathematical Preliminaries:

Suppose that $A = A^T$ and $A = Q\Lambda Q^T$ with eigenvalues sorted so $\lambda_1 \geq \cdots \geq \lambda_n$. Then,

$$\begin{aligned}x^T A x &= x^T Q \Lambda Q^T x \\&= (Q^T x)^T \Lambda (Q^T x) \\&= \sum_{i=1}^n \lambda_i (q_i^T x)^2 \\&\leq \lambda_1 \sum_{i=1}^n (q_i^T x)^2 = \lambda_1 [(Q^T x)^T I (Q^T x)] = \lambda_1 x^T x \\&= \lambda_1 \|x\|^2.\end{aligned}$$

Mathematical Preliminaries:

- That is, we have $x^T A x \leq \lambda_1 x^T x$.
- By a similar argument, we can find that $x^T A x \geq \lambda_n \|x\|^2$, so we have

$$\lambda_n x^T x \leq x^T A x \leq \lambda_1 x^T x.$$

- Note also that $q_1^T A q_1 = \lambda_1 \|q_1\|^2$ and $q_n^T A q_n = \lambda_n \|q_n\|^2$, so the inequalities are tight.

Mathematical Preliminaries:

Matrix gain:

- Suppose now that $A \in \mathbb{R}^{m \times n}$ (not necessarily square or symmetric).
- For $x \in \mathbb{R}^n$, $\|Ax\|/\|x\|$ gives the amplification factor or gain of A in the direction x .
- The max. gain is called the matrix norm of A , and is denoted $\|A\|$.

Mathematical Preliminaries:

- The matrix p -norm is defined as

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}.$$

- For $p = 2$,

$$\sup_{x \neq 0} \frac{\|Ax\|_2^2}{\|x\|_2^2} = \sup_{x \neq 0} \frac{x^T A^T A x}{\|x\|_2^2} = \lambda_{\max}(A^T A),$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}.$$

Mathematical Preliminaries:

- The min. gain is given by

$$\inf_{x \neq 0} \|Ax\|_2 / \|x\|_2 = \sqrt{\lambda_{\min}(A^T A)}.$$

- Note that $A^T A \in \mathbb{R}^{n \times n}$ is symmetric and $A^T A \geq 0$ so $\lambda_{\min}$ and $\lambda_{\max} \geq 0$.
- The “max gain” input direction is $x = q_1$, the eigenvector of $A^T A$ associated with $\lambda_{\max}$.

Mathematical Preliminaries:

- The “min gain” input direction is $x = q_n$, the eigenvector of $A^T A$ associated with $\lambda_{\min}$.

- For $p = 1$
$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|,$$

- For $p = \infty$
$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|,$$

Mathematical Preliminaries:

Singular Value Decomposition: Finding matrix 1- and ∞ -norms is quite simple. Finding the 2-norm is more difficult because of the need to find eigenvalues, eigenvectors.

We can replace this step with one that is computationally simpler by using the singular value decomposition (SVD) of $A \in \mathbb{R}^{m \times n}$, where $\text{rank}(A) = r$:

Mathematical Preliminaries:

SVD

$$A = U\Sigma V^T.$$

- $U = [u_1, \dots, u_r] \in \mathbb{R}^{m \times r}$, and $U^T U = I$, and u_i are the left or output singular vectors of A .
- $V = [v_1, \dots, v_r] \in \mathbb{R}^{n \times r}$, and $V^T V = I$, and v_i are the right or input singular vectors of A .
- $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ where $\sigma_1 \geq \dots \geq \sigma_r > 0$, and σ_i are the (nonzero) singular values of A .

$$\sigma_i = \sqrt{\lambda_i(A^T A)} \text{ and hence } \|A\|_2 = \sigma_1.$$

Mathematical Preliminaries:

Cayley Hamilton Theorem: A square matrix A satisfies its own characteristic equation. That is, if $\chi(\lambda) = \det(\lambda I - A) = 0$ then $\chi(A) = 0$. We can easily show this if A is diagonalizable. Let $A = V^{-1}\Lambda V$. Then,

$$A^2 = V^{-1}\Lambda V V^{-1}\Lambda V$$

$$= V^{-1}\Lambda^2 V$$

$$A^k = V^{-1}\Lambda^k V.$$

Mathematical Preliminaries:

The characteristic polynomial is:

$$\chi(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1$$

so if we replace λ with A we get

$$\begin{aligned}\chi(A) &= A^n + a_{n-1}A^{n-1} + \cdots + a_1I \\ &= V^{-1} [\Lambda^n + a_{n-1}\Lambda^{n-1} + \cdots + a_1I] V.\end{aligned}$$

- To “prove” the Cayley–Hamilton theorem, we just need to show that the quantity inside the brackets is zero.

Mathematical Preliminaries:

- It is a diagonal matrix, and each diagonal element has the form

$$\lambda_i^n + a_{n-1}\lambda_i^{n-1} + \cdots + a_1 = 0$$

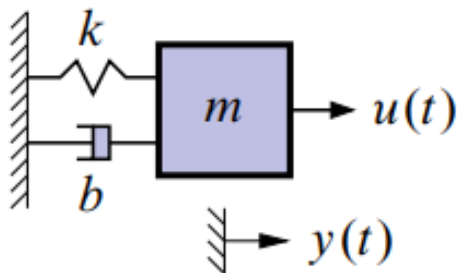
because λ_i is an eigenvalue of A .

- So, each diagonal element is zero.

State-Space Representation:

Representation of dynamics of an n th-order system as a first-order differential equation in an n -vector called the state $\Rightarrow$ n first-order equations.

Example: 2nd-order equation of motion (EOM).



$$m\ddot{y}(t) = u(t) - b\dot{y}(t) - ky(t)$$
$$\Rightarrow \ddot{y}(t) = \frac{u(t) - b\dot{y}(t) - ky(t)}{m}$$

State-Space Representation:

Define state vector.

$$x(t) = \begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix}$$

$$\text{then, } \dot{x}(t) = \begin{bmatrix} \dot{y}(t) \\ \ddot{y}(t) \end{bmatrix} = \begin{bmatrix} \dot{y}(t) \\ -\frac{k}{m}y(t) - \frac{b}{m}\dot{y}(t) + \frac{1}{m}u(t) \end{bmatrix}.$$

State-Space Representation:

We can write this in the form

$\dot{x}(t) = Ax(t) + Bu(t)$ where A and B are constant matrices:

$$\dot{x}(t) = \begin{bmatrix} \dot{y}(t) \\ \ddot{y}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} \quad \\ \quad \end{bmatrix}}_A \begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix} + \underbrace{\begin{bmatrix} \quad \\ \quad \end{bmatrix}}_B u(t).$$

State-Space Representation:

Complete picture by finding $y(t)$ as function of $x(t)$: General form is

$$y(t) = Cx(t) + Du(t),$$

where C and D are constant matrices.

State-Space Representation:

- Fundamental form for LTI state-space model:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t),$$

where $u(t)$ is the input, $x(t)$ is the “state”, A , B , C , D are constant matrices.

State-Space Representation:

■ **DEFINITION:** The state of a system at time t_0 is the minimum amount of information at t_0 that, together with the input $u(t)$, $t \geq t_0$, uniquely determines the behavior of the system for all $t \geq t_0$.

Contrast with impulse-response (convolution) representation which requires all past history of $u(t)$

$$y(t) = \int_0^t h(\tau)u(t - \tau) d\tau.$$

State-Space Representation:

- Transfer functions provide input-output mapping: $u \mapsto G(s) \mapsto y$. State variables provide access to what is going on inside the system.
- Convenient way to express EOM. Matrix format great for computers.
- Allows new analysis and synthesis tools.
- Great for multi-input, multi-output systems. These are very hard to work with transfer functions.

State-Space to Transfer Function:

State-space model:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t),$$

Laplace Transform:

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s),$$

State-Space to Transfer Function:

$$(sI - A)X(s) = BU(s) + x(0)$$

$$X(s) = (sI - A)^{-1}BU(s) + (sI - A)^{-1}x(0)$$

$$Y(s) = \underbrace{[C(sI - A)^{-1}B + D]}_{\text{transfer function of system}} U(s) + \underbrace{C(sI - A)^{-1}x(0)}_{\text{response to initial conditions}} .$$

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D,$$

State-Space to Transfer Function:

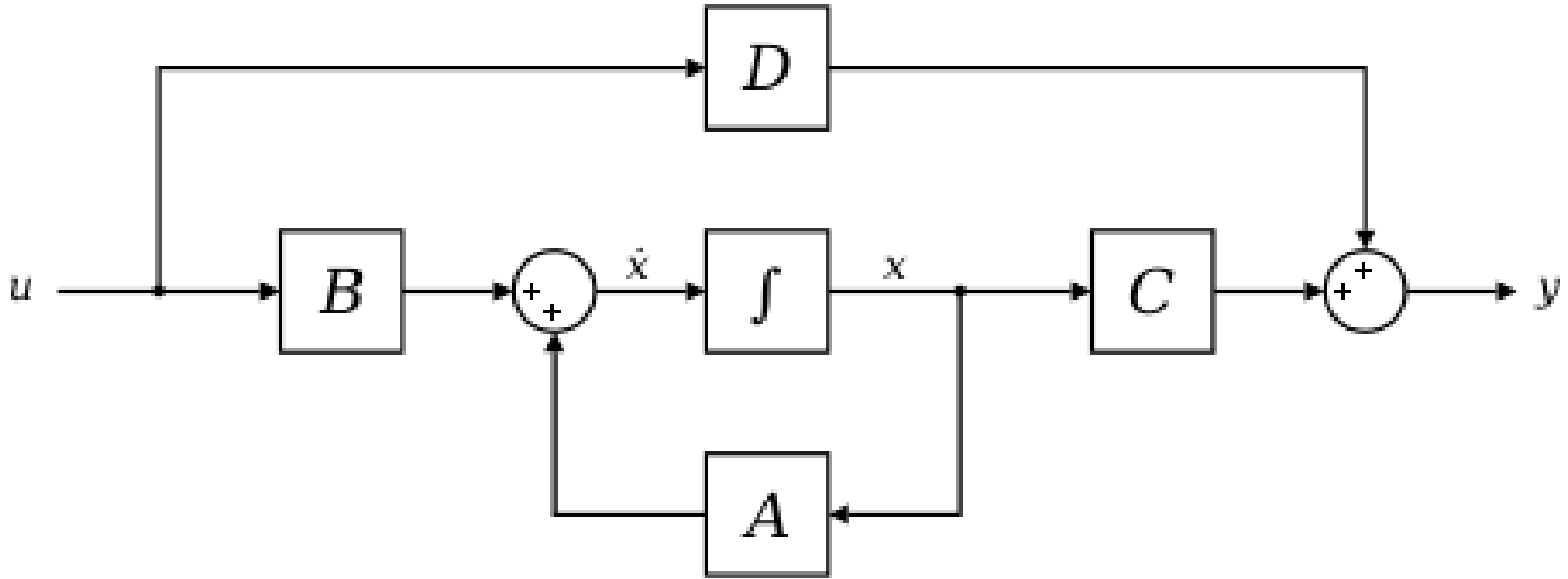
Example: Consider mass-spring-damper example.

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The transfer function is

$$\begin{aligned} G(s) &= C(sI - A)^{-1}B + 0 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s & -1 \\ \frac{k}{m} & s + \frac{b}{m} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} \\ &= \frac{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s + \frac{b}{m} & 1 \\ -\frac{k}{m} & s \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}{s^2 + (b/m)s + (k/m)} = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}} = \frac{1}{ms^2 + bs + k}. \end{aligned}$$

State-Space to Transfer Function:



The block diagram representation of linear state-space model

Canonical Forms:

Controller Canonical Form:

Case 1 : Transfer function is made up only of poles.

$$G(s) = \frac{1}{s^3 + a_1s^2 + a_2s + a_3} = \frac{Y(s)}{U(s)}$$

$$\Rightarrow \ddot{y}(t) + a_1\ddot{y}(t) + a_2\dot{y}(t) + a_3y(t) = u(t).$$

Canonical Forms:

Controller Canonical Form: Choose output and derivatives as the state.

$x(t) = \begin{bmatrix} \ddot{y}(t) & \dot{y}(t) & y(t) \end{bmatrix}^T$. Then

$$\dot{x}(t) = \begin{bmatrix} \ddot{y}(t) \\ \dot{y}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \ddot{y}(t) \\ \dot{y}(t) \\ y(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t).$$

A is a top-companion matrix.

Canonical Forms:

Controller Canonical Form:

Case 2 : Transfer function has poles and zeros, but is strictly proper.

$$G(s) = \frac{b_1s^2 + b_2s + b_3}{s^3 + a_1s^2 + a_2s + a_3} = \frac{Y(s)}{U(s)}.$$

Break up transfer function into two parts.

$\frac{V(s)}{U(s)}$ contains all of the poles of $\frac{Y(s)}{U(s)}$.

$$Y(s) = [b_1s^2 + b_2s + b_3]V(s).$$

Canonical Forms:

Controller Canonical Form:

$$y(t) = b_1 \ddot{v}(t) + b_2 \dot{v}(t) + b_3 v(t).$$

$$V(s)[s^3 + a_1 s^2 + a_2 s + a_3] = U(s),$$

$$\ddot{v}(t) + a_1 \dot{v}(t) + a_2 v(t) + a_3 v(t) = u(t).$$

Canonical Forms:

Controller Canonical Form:

$$\dot{x}(t) = \begin{bmatrix} \ddot{v}(t) \\ \ddot{v}(t) \\ \dot{v}(t) \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \ddot{v}(t) \\ \dot{v}(t) \\ v(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$Y(s) = [b_1 s^2 + b_2 s + b_3] V(s)$$

$$y(t) = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t).$$

Canonical Forms:

Controller Canonical Form:

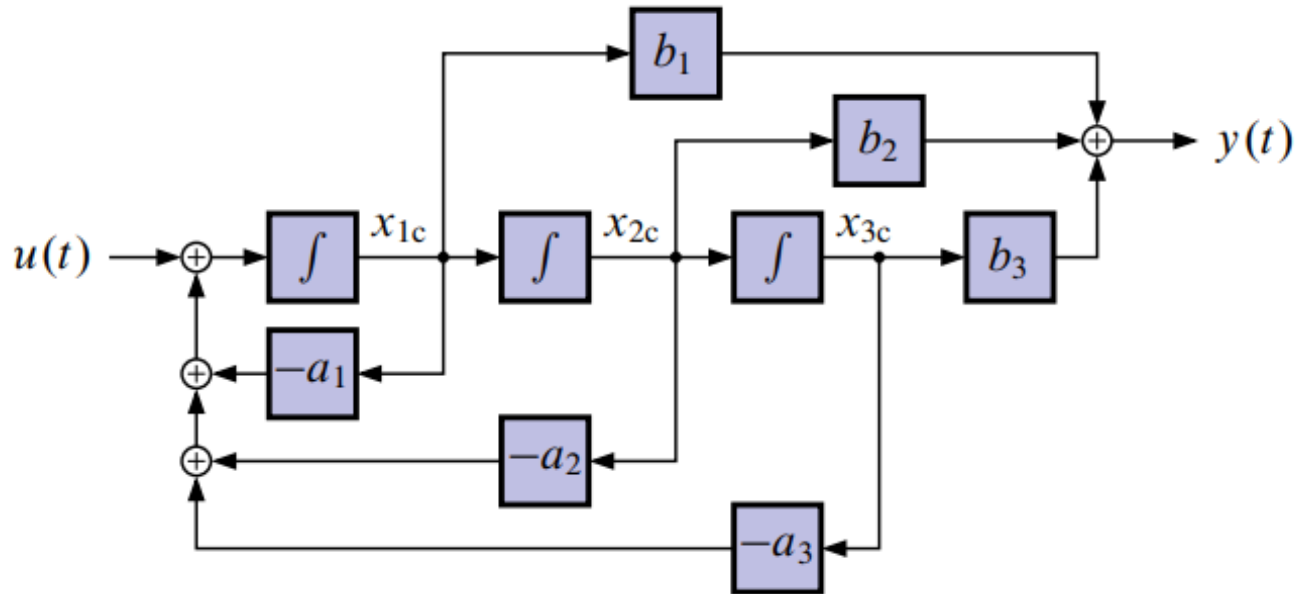
Case 3 : Non-proper transfer function.

$$\begin{aligned} G(s) &= \frac{b_0s^3 + b_1s^2 + b_2s + b_1}{s^3 + a_1s^2 + a_2s + a_3} \\ &= \frac{\beta_1s^2 + \beta_2s + \beta_3}{s^3 + a_1s^2 + a_2s + a_3} + D, \end{aligned}$$

where the β_i terms are computed via long division. The remainder D is the feedthrough term.

Canonical Forms:

Controller Canonical Form:



Block diagram of controller canonical form

Canonical Forms:

Observer Canonical Form:

$$G(s) = \frac{b_1s^2 + b_2s + b_3}{s^3 + a_1s^2 + a_2s + a_3} = \frac{Y(s)}{U(s)}.$$

$$(s^3 + a_1s^2 + a_2s + a_3)Y(s) = (b_1s^2 + b_2s + b_3)U(s),$$

Divide both side by s^3

$$\begin{aligned} Y(s) &= \left(-\frac{a_1}{s} - \frac{a_2}{s^2} - \frac{a_3}{s^3}\right) Y(s) + \left(\frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3}\right) U(s) \\ &= \frac{1}{s} \left(b_1 U(s) - a_1 Y(s) + \frac{1}{s} \left(b_2 U(s) - a_2 Y(s) + \frac{1}{s} \left(b_3 U(s) - a_3 Y(s) \right) \right) \right). \end{aligned}$$

Canonical Forms:

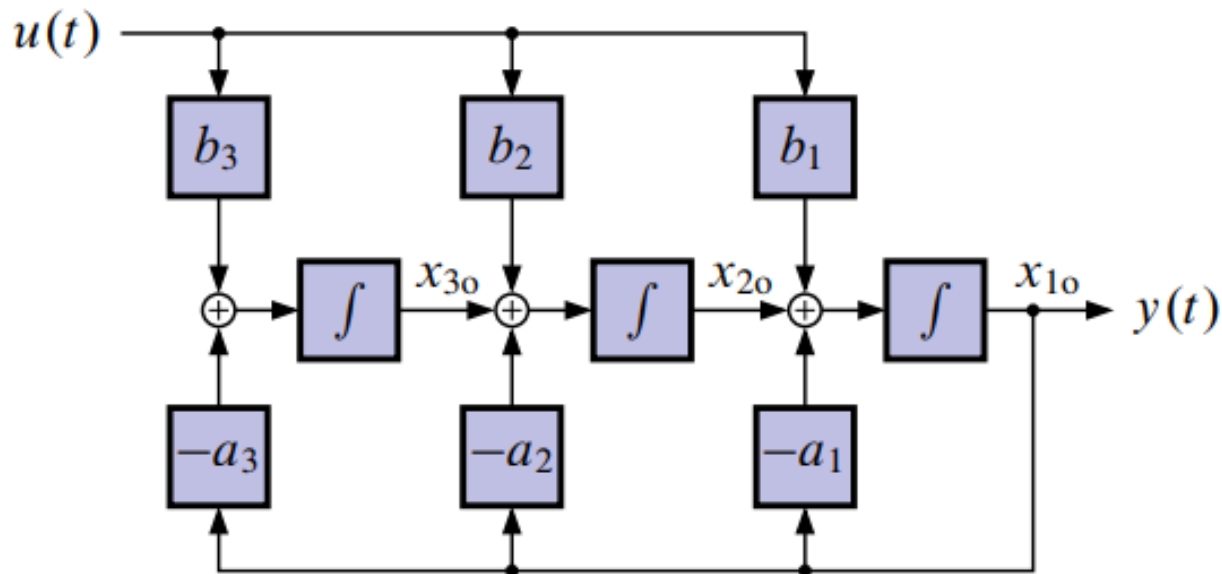
Observer Canonical Form:

$$\dot{x}(t) = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t).$$

A is a left-companion matrix.

Canonical Forms:

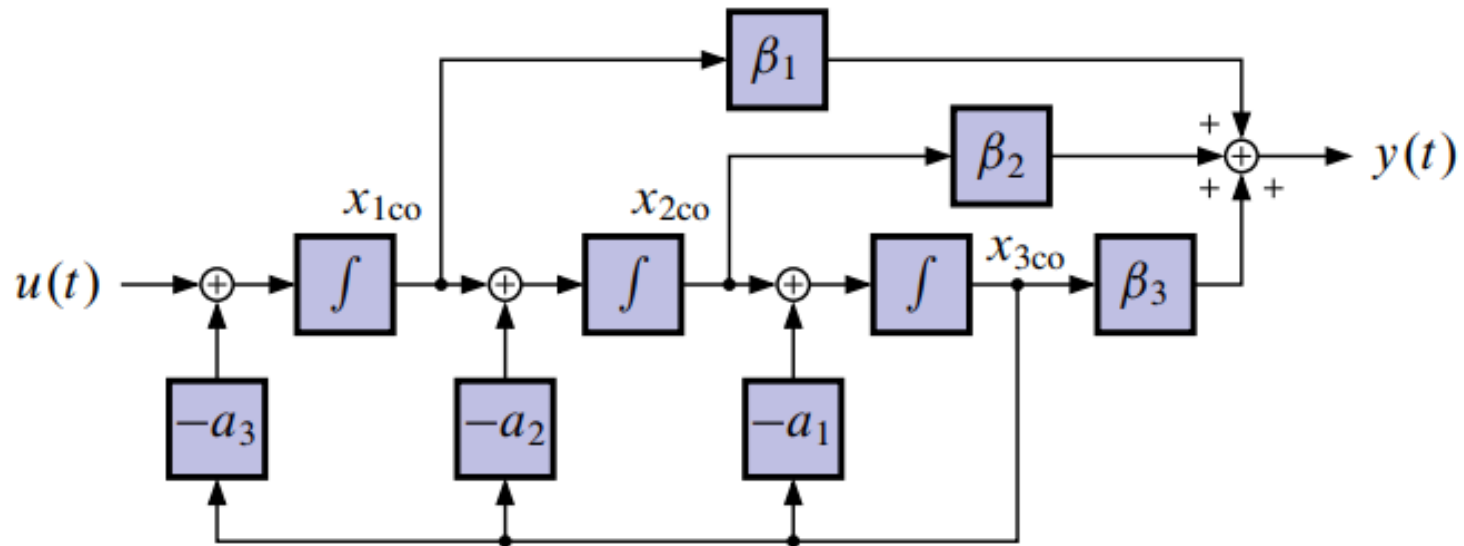
Observer Canonical Form:



Block diagram of observer canonical form

Canonical Forms:

Controllability Canonical Form:



Block diagram of controllability canonical form

Canonical Forms:

Controllability Canonical Form:

$$\begin{aligned}x_3 &= \frac{1}{s}(x_2 - a_1x_3) \\x_2 &= \frac{1}{s}(x_1 - a_2x_3) \\x_1 &= \frac{1}{s}(u - a_3x_3)\end{aligned} \quad \Rightarrow \quad \begin{aligned}X_3(s) &= \frac{1}{s^3 + a_1s^2 + a_2s + a_3}U(s) \\X_2(s) &= \frac{s + a_1}{s^3 + a_1s^2 + a_2s + a_3}U(s) \\X_1(s) &= \frac{s^2 + a_1s + a_2}{s^3 + a_1s^2 + a_2s + a_3}U(s).\end{aligned}$$

Thus,

$$Y(s) = \frac{\beta_3 + \beta_2(s + a_1) + \beta_1(s^2 + a_1s + a_2)}{s^3 + a_1s^2 + a_2s + a_3}U(s).$$

Canonical Forms:

Controllability Canonical Form:

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a_1 & 1 & 0 \\ a_2 & a_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 - a_1 b_1 \\ b_3 - a_1 b_2 - a_2 b_1 + a_1^2 b_1 \end{bmatrix}.$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 0 & -a_3 \\ 1 & 0 & -a_2 \\ 0 & 1 & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} \beta_1 & \beta_2 & \beta_3 \end{bmatrix} x(t).$$

A is a right-companion matrix.

Canonical Forms:

Observability Canonical Form:

- Note that $H(s)$ is a scalar.

So, $H^T(s) = H(s)$.

$$\begin{aligned} H(s) &= C(sI - A)^{-1}B + D \\ &= B^T(sI - A)^{-T}C^T + D^T \\ &= B^T(sI - A^T)^{-1}C^T + D^T. \end{aligned}$$

- So, $C \leftrightarrow B^T$, $A \leftrightarrow A^T$, $B \leftrightarrow C^T$
and $D \leftrightarrow D^T$ are **dual** forms.

Canonical Forms:

Observability Canonical Form:

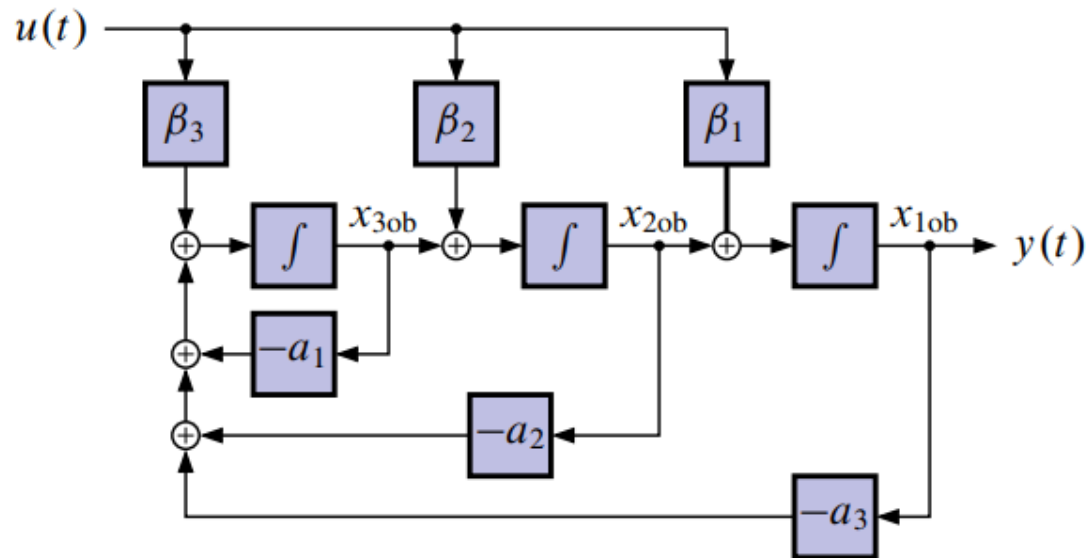
$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t)$$

Observability canonical form is a dual form with the controllability canonical form.

A is a bottom companion matrix.

Canonical Forms:

Observability Canonical Form:



Block diagram of observability canonical form

Modal (Diagonal) Form:

- Assume $G(s) = \frac{N(s)}{D(s)}$.
- $D(s)$ has distinct roots p_i (real).

$$\begin{aligned} G(s) &= \frac{N(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)} \\ &= \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n}. \end{aligned}$$

$$\begin{aligned} \frac{X_1(s)}{U(s)} &= \frac{r_1}{s - p_1} \quad \Rightarrow \quad \dot{x}_1(t) = p_1 x_1(t) + r_1 u(t) \\ &\vdots \\ \frac{X_n(s)}{U(s)} &= \frac{r_n}{s - p_n} \quad \Rightarrow \quad \dot{x}_n(t) = p_n x_n(t) + r_n u(t). \end{aligned}$$

Modal (Diagonal) Form:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

$$A = \begin{bmatrix} p_1 & & & 0 \\ & p_2 & & \\ & & \ddots & \\ 0 & & & p_n \end{bmatrix}, \quad B = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \end{bmatrix}.$$

- Easily extend to handle complex poles $\lambda_i = \sigma_i + j\omega_i$.

Modal (Diagonal) Form:

- If A and B may have complex elements, no change is necessary.
- Otherwise, use “real modal form” which is made via partial-fraction expansion where complex pole-pairs are represented as

$$G_i(s) = \frac{\alpha_i s + \beta_i}{(s - \sigma_i)^2 + \omega_i^2}.$$

- The real-modal form has an A matrix which is block diagonal, and of the form

$$A = \text{diag} \left(\Lambda_r, \begin{bmatrix} \sigma_{r+1} & \omega_{r+1} \\ -\omega_{r+1} & \sigma_{r+1} \end{bmatrix}, \dots, \begin{bmatrix} \sigma_n & \omega_n \\ -\omega_n & \sigma_n \end{bmatrix} \right)$$

where Λ_r is a diagonal matrix containing the real poles, and

$$\lambda_i = \sigma_i + j\omega_i, \quad i = r + 1, \dots, n$$

are the complex poles.

Modal (Diagonal) Form:

- The B matrix has corresponding entries:

$$\begin{bmatrix} b_{i,1} \\ b_{i,2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -\omega_i - \sigma_i & \omega_i - \sigma_i \end{bmatrix}^{-1} \begin{bmatrix} \alpha_i \\ \beta_i \end{bmatrix}.$$

- Modal form is convenient for keeping track of system poles. . . they are right on the diagonal!
- Good representation to use. . . numerical robustness.
- *All canonical forms related by linear algebra—change of basis.*
- Diagonal is very useful, but we cannot always put a system in diagonal form. (What was our assumption above?)
- We will see one more canonical (Jordan) form in a little while that is very similar to diagonal. *All* systems can be put in Jordan form.